Mina Magdy

SOFTWARE AI ENGINEER

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Full-time Software AI Engineer at QovaTech, building AI-powered business tools and backend services. Built a bilingual Arabic and English retrieval system, achieving Recall@5 of 1.00 and MRR@5 of 0.80 in both languages.

EXPERIENCE

QovaTech | Software AI Engineer | Full-time Jan 2026 - Present
Alexandria, Egypt | Hybrid
- Build AI-powered tools and integrate language models and third-party services into business application workflows.
- Develop backend services, APIs, data models, and workflow logic for CRM and business applications.
- Implement access controls, database transactions, validation, caching, and automated backend tests.

QovaTech | Software Engineer Intern Nov 2025 - Jan 2026
- Contributed to backend development and AI integrations before transitioning to a full-time role.

TechXlab | Software Engineering Intern Jul 2025 - Sep 2025
Vodafone-sponsored internship
- Developed a Java backend project in an Agile/Scrum team and earned an A+ internship grade.

SELECTED PROJECTS

Client Worklog AI | Personal project 2026
Python, Sentence Transformers, Chroma, PostgreSQL, FastAPI, React
- Built text, image, and audio evidence pipelines that preserve source provenance before work-ledger updates.
- Implemented project-scoped RAG with SQL revalidation, citation allowlists, and a revision-aware LRU/TTL cache; used SQL to keep monthly reports complete.
- Evaluated 20 synthetic Arabic and English queries, achieving Recall@5 of 1.00 and MRR@5 of 0.80 for each language; measured warm retrieval latency and memory use.
- Built async FastAPI services and React review, search, and reporting views, with Alembic migrations and mocked/PostgreSQL integration tests.

PostPilot | AI Content Platform 2026 | QovaTech project
NestJS, PostgreSQL, Redis | Private repository
- Built semantic caching to reduce redundant inference and a prompt regression harness to validate output quality before prompt changes.

AI Logger | AI-Assisted Observability GitHub project
NestJS, TypeScript, SQLite
- Integrated interchangeable language model providers for batch log analysis, health summaries, and root-cause investigation using surrounding log context.
- Built REST, Syslog, and SSH ingestion with event-driven incident workflows and Slack, webhook, and email notifications.

TECHNICAL SKILLS

Machine learning frameworks: PyTorch, scikit-learn
LLM applications and agents: LangGraph, AI agents, MCP, prompt engineering, structured outputs
RAG and evaluation: Sentence Transformers, multilingual embeddings, Chroma, semantic search, Recall@5, MRR@5, prompt regression tests
Programming and APIs: Python, SQL, TypeScript, FastAPI, NestJS, Pydantic, React, REST APIs, asynchronous programming
Data and delivery: PostgreSQL, Redis, Docker, AWS, Linux, Git, CI/CD

EDUCATION

Alexandria University | B.Sc. in Computer Science Sep 2023 - Jul 2027
Faculty of Computer and Data Science | Expected graduation: July 2027 | Grade: Excellent
Relevant coursework: Machine Learning, Linear Algebra, Statistics and Probability, Algorithms, Database Systems.

LEADERSHIP & LANGUAGES

Project Manager, IEEE Xtreme | IEEE Alex Computer Science Chapter | Apr 2024 - Apr 2025
Organized the IEEE Xtreme competition. Solved 700+ problems on Codeforces and LeetCode. **Languages:** Arabic (Native), English (Proficient)